

Bayesian Governance: A Mathematical Framework for Continuous Audit Monitoring

A Review

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Abstract

Traditional audit methodologies rely on binary pass/fail judgments that discard valuable information about the *degree* of control effectiveness and its evolution over time. This review paper presents a unified mathematical framework for *continuous governance monitoring* grounded in Bayesian inference, information theory, and geometric statistics. We model control effectiveness as a latent parameter $\theta \in [0, 1]$ governed by a Beta-Binomial conjugate family, updated sequentially with each new audit observation. Shannon entropy of the posterior serves as a natural early-warning signal for governance deterioration, while Kullback–Leibler and Jensen–Shannon divergences enable formal regime-change detection. Monte Carlo stress testing propagates posterior uncertainty through correlated control portfolios to produce principled Value-at-Risk estimates. Finally, Jensen’s inequality reveals that ignoring variance in control effectiveness systematically *underestimates* expected risk—a result with direct implications for audit planning under ISA 315. We survey the relevant mathematical foundations, present illustrative computational experiments, and outline a research agenda connecting Bayesian governance to topological data analysis and information geometry for regulated industries.

Keywords: Beta-Binomial models, Bayesian inference, Shannon entropy, Monte Carlo simulation, governance risk, continuous auditing, information geometry, ISA 315.

Contents

1	Introduction	2
2	Bayesian Foundations: The Beta-Binomial Model	3
2.1	Setup and Notation	3
2.2	Conjugate Updating	3
2.3	Posterior Summaries for Audit Reporting	4
3	Shannon Entropy as Early-Warning Signal	4
4	Information-Geometric Regime Detection	5
4.1	Kullback–Leibler Divergence	5
4.2	Jensen–Shannon Divergence	5
4.3	Connection to Fisher–Rao Geometry	6

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5 Monte Carlo Stress Testing	6
5.1 Portfolio Risk Model	6
5.2 Simulation Algorithm	7
5.3 Implications for Audit Planning	7
6 Jensen’s Inequality and the Variance Penalty	7
7 Integrated Framework Architecture	8
8 Connections to Information Geometry and TDA	9
8.1 Information Geometry (Paper A)	9
8.2 Topological Data Analysis (Papers A and C)	10
8.3 Entity-Level Analysis (Paper B)	10
9 Discussion and Research Agenda	10
9.1 Summary of Contributions	10
9.2 Limitations	11
9.3 Open Problems and Future Directions	11
9.4 Concluding Remarks	11

1 Introduction

The audit profession faces a fundamental tension. International Standards on Auditing (ISA 315, ISA 300) demand that auditors assess internal controls, identify risks of material misstatement, and plan audit procedures accordingly. Yet the dominant paradigm remains one of *point-in-time* assessment: controls are tested, classified as effective or deficient, and the finding is recorded in a static report. This binary paradigm discards two critical dimensions of information:

1. **Uncertainty:** A control tested five times with four passes is not the same as one tested fifty times with forty passes, yet both yield an 80% pass rate.
2. **Dynamics:** A control whose effectiveness is declining from 90% to 60% over twelve months presents a qualitatively different risk profile than one that has been stable at 75%.

A broader research programme¹ addresses this gap by bringing the mathematical apparatus of Bayesian statistics, information theory, and differential geometry to bear on audit and governance monitoring. This review paper focuses on the *Bayesian governance* component: a framework that replaces binary judgments with calibrated probability distributions over control effectiveness, updated continuously with each new observation.

The approach is not merely theoretical. In regulated industries such as pharmaceuticals—where the EU AI Act (effective August 2026 for high-risk systems) demands ongoing conformity assessment of AI systems—the ability to maintain *continuous, quantitative assurance* that controls are working is rapidly becoming a regulatory necessity, not an academic luxury.

The remainder of this paper is organized as follows. Section 2 establishes the Beta-Binomial model and its conjugate updating mechanics. Section 3 introduces Shannon entropy as an early-warning signal. Section 4 develops KL and JS divergence for regime detection. Section 5 extends the framework to portfolio-level stress testing. Section 6 presents the Jensen’s inequality result on risk underestimation. Section 7 describes the integrated framework architecture. Section 8 surveys connections to information geometry and topological data analysis. Section 9 concludes with a research agenda.

¹The programme comprises three papers: Paper A (geometric theory), Paper B (entity-level failure analysis), and Paper C (process-level domain features).

2 Bayesian Foundations: The Beta-Binomial Model

2.1 Setup and Notation

Let $\theta \in [0, 1]$ denote the *true* effectiveness of an internal control—the probability that, on any given test, the control operates as designed. We treat θ as a latent random variable and place a Beta prior on it:

Definition 2.1 (Beta Prior). The prior belief about control effectiveness is modelled as

$$\theta \sim \text{Beta}(\alpha_0, \beta_0), \quad p(\theta) = \frac{\theta^{\alpha_0-1}(1-\theta)^{\beta_0-1}}{B(\alpha_0, \beta_0)}, \quad (1)$$

where $B(\alpha_0, \beta_0) = \frac{\Gamma(\alpha_0)\Gamma(\beta_0)}{\Gamma(\alpha_0+\beta_0)}$ is the Beta function, and $\alpha_0, \beta_0 > 0$ encode prior pseudo-counts of successes and failures respectively.

The choice of $\alpha_0 = \beta_0 = 2$ represents a mildly informative prior centred at $\theta = 0.5$ with moderate uncertainty—a reasonable default for a control about which the auditor has no prior testing history.

2.2 Conjugate Updating

Each audit test of the control yields a binary outcome $x_t \in \{0, 1\}$ (fail or pass). Given n tests with s successes, the likelihood is Binomial:

$$p(\mathbf{x} | \theta) = \binom{n}{s} \theta^s (1 - \theta)^{n-s}. \quad (2)$$

By conjugacy, the posterior is again Beta:

$$\theta | \mathbf{x} \sim \text{Beta}(\alpha_0 + s, \beta_0 + n - s). \quad (3)$$

This is the core computational engine of the framework. Each new observation x_t updates the posterior in constant time: $\alpha \leftarrow \alpha + x_t$, $\beta \leftarrow \beta + (1 - x_t)$. No matrix inversions, no MCMC chains—just increment two counters. Figure 1 illustrates the sequential narrowing of the posterior as evidence accumulates, and the shift that occurs when control effectiveness deteriorates.

Figure 1: Sequential Bayesian Posterior Update for Control Effectiveness

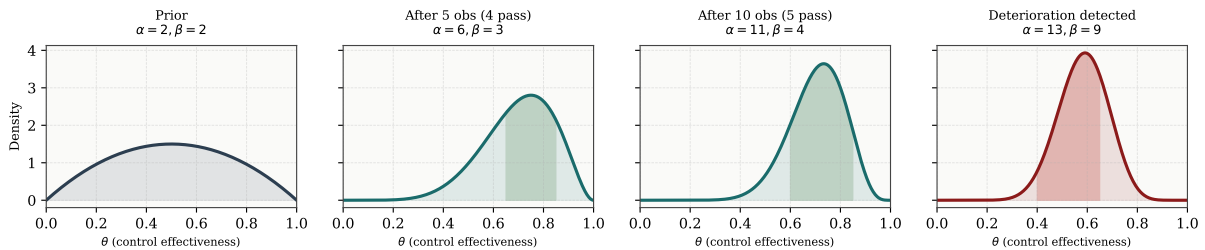


Figure 1: Sequential Bayesian posterior update for a single control. The prior (far left) is updated with favourable observations (centre panels), then deterioration produces a leftward shift and increased entropy (far right). Shaded regions indicate 80% highest-density intervals.

2.3 Posterior Summaries for Audit Reporting

The posterior $\text{Beta}(\alpha, \beta)$ yields several quantities directly useful for audit judgment:

- **Point estimate:** $\hat{\theta} = \frac{\alpha}{\alpha + \beta}$ (posterior mean).
- **Uncertainty:** $\text{Var}[\theta] = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}$.
- **Credible interval:** The 95% highest-density interval (HDI) provides a rigorous analogue of the confidence interval.
- **Probability of adequacy:** $\Pr(\theta > \theta^*) = 1 - I_{\theta^*}(\alpha, \beta)$, where I_x is the regularized incomplete Beta function.

The last quantity is particularly powerful: rather than declaring a control “effective” or “deficient,” the auditor can report, for instance, that there is an 87% posterior probability that the control’s true effectiveness exceeds the 70% threshold required by internal policy.

3 Shannon Entropy as Early-Warning Signal

While the posterior mean $\hat{\theta}$ tracks the *level* of control effectiveness, it does not capture the *regime uncertainty* that often precedes governance failure. Shannon entropy provides this missing signal.

Definition 3.1 (Differential Entropy of a Beta Distribution). For $\theta \sim \text{Beta}(\alpha, \beta)$, the differential entropy is

$$H[\theta] = \ln B(\alpha, \beta) - (\alpha - 1)\psi(\alpha) - (\beta - 1)\psi(\beta) + (\alpha + \beta - 2)\psi(\alpha + \beta), \quad (4)$$

where $\psi(\cdot)$ is the digamma function.

Proposition 3.1 (Entropy as Deterioration Detector). Consider a control monitored sequentially. If the true effectiveness θ^* shifts from a high-effectiveness regime ($\theta^* \approx 0.85$) to a deteriorated regime ($\theta^* \approx 0.50$), the posterior entropy $H[\theta_t]$ exhibits a transient *increase* before the posterior mean $\hat{\theta}_t$ has shifted appreciably. This entropy spike serves as a leading indicator of governance deterioration.

The intuition is straightforward: when the data-generating regime changes, the posterior must “unlearn” its previous concentration, producing a temporary broadening (entropy increase) before re-concentrating around the new regime. Figure 2 demonstrates this phenomenon computationally.

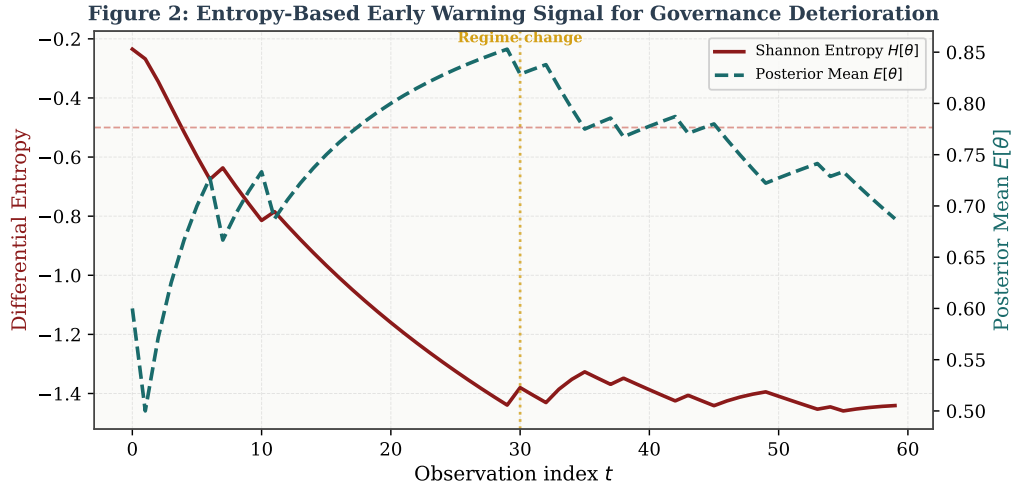


Figure 2: Shannon entropy (solid) and posterior mean (dashed) over 60 sequential observations. A regime change at $t = 30$ produces an entropy signal that leads the mean shift by several observations, providing an early-warning window for governance intervention.

Remark 3.1. In practice, the entropy signal can be monitored against control limits (analogous to statistical process control charts). An entropy exceedance triggers deeper investigation, not an automatic finding—preserving the auditor’s professional judgment while providing a principled quantitative trigger.

4 Information-Geometric Regime Detection

While entropy monitors the *absolute* uncertainty of a single posterior, divergence measures quantify the *distance* between posteriors at different times, enabling formal regime-change detection.

4.1 Kullback–Leibler Divergence

The KL divergence between the current posterior $P_t = \text{Beta}(\alpha_t, \beta_t)$ and a reference posterior $P_{t-w} = \text{Beta}(\alpha_{t-w}, \beta_{t-w})$ is:

$$D_{\text{KL}}(P_t \| P_{t-w}) = \ln \frac{B(\alpha_{t-w}, \beta_{t-w})}{B(\alpha_t, \beta_t)} + (\alpha_t - \alpha_{t-w})\psi(\alpha_t) + (\beta_t - \beta_{t-w})\psi(\beta_t) + (\alpha_{t-w} - \alpha_t + \beta_{t-w} - \beta_t)\psi(\alpha_t + \beta_t). \quad (5)$$

4.2 Jensen–Shannon Divergence

Since D_{KL} is asymmetric and unbounded, the symmetric Jensen–Shannon divergence is often preferred:

$$D_{\text{JS}}(P_t \| P_{t-w}) = \frac{1}{2} D_{\text{KL}}(P_t \| M) + \frac{1}{2} D_{\text{KL}}(P_{t-w} \| M), \quad (6)$$

where $M = \frac{1}{2}(P_t + P_{t-w})$ is the mixture distribution. $D_{\text{JS}} \in [0, \ln 2]$ and its square root is a proper metric on the space of probability distributions.

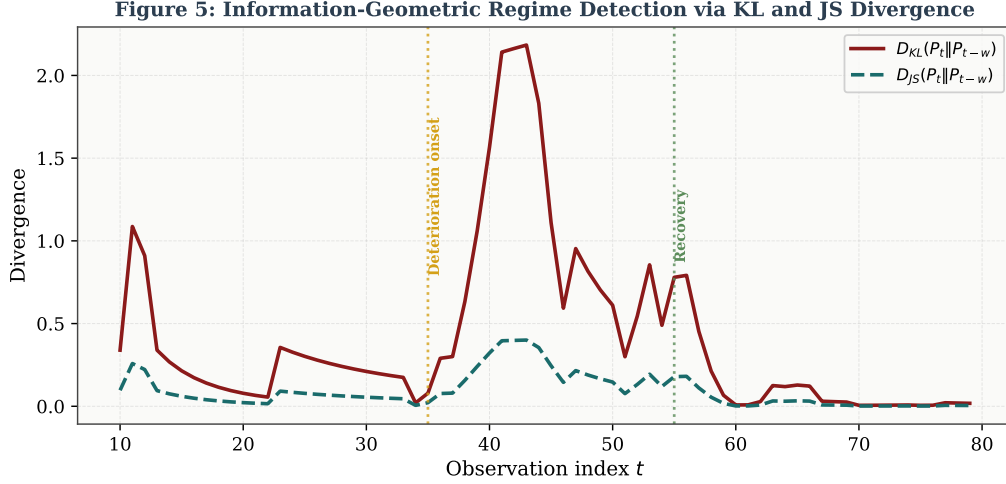


Figure 3: KL and JS divergence between the current posterior and a lagged reference ($w = 10$). Divergence spikes at the onset of deterioration ($t = 35$) and again at recovery ($t = 55$), enabling automated regime-change detection with controlled false-positive rates.

4.3 Connection to Fisher–Rao Geometry

The Beta family forms a two-dimensional statistical manifold $\mathcal{M} = \{(\alpha, \beta) : \alpha, \beta > 0\}$ equipped with the Fisher information metric:

$$g_{ij}(\alpha, \beta) = \begin{pmatrix} \psi_1(\alpha) - \psi_1(\alpha + \beta) & -\psi_1(\alpha + \beta) \\ -\psi_1(\alpha + \beta) & \psi_1(\beta) - \psi_1(\alpha + \beta) \end{pmatrix}, \quad (7)$$

where ψ_1 is the trigamma function. The geodesic distance on this manifold—the Fisher–Rao distance—provides the *intrinsic* measure of how far two posteriors are in the space of all possible beliefs about control effectiveness. This geometric perspective, developed fully in Paper A, reveals that governance deterioration corresponds to *curvature changes* on the statistical manifold, detectable via persistent homology.

5 Monte Carlo Stress Testing

A single control’s posterior is useful; a *portfolio* of control posteriors is transformative. Organizations maintain hundreds of controls, and the joint failure distribution—not just individual pass rates—determines enterprise risk.

5.1 Portfolio Risk Model

Let $\theta_1, \dots, \theta_K$ be the effectiveness parameters for K controls, each with posterior $\theta_k \sim \text{Beta}(\alpha_k, \beta_k)$. Define the portfolio effectiveness score:

$$\Theta = \frac{1}{K} \sum_{k=1}^K w_k \theta_k, \quad (8)$$

where w_k are importance weights reflecting the control’s relevance to the audit objective.

5.2 Simulation Algorithm

Algorithm 1 Monte Carlo Stress Test for Control Portfolio

Require: Posteriors $\{(\alpha_k, \beta_k)\}_{k=1}^K$, weights $\{w_k\}$, simulations N , threshold θ^*

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1: for  $i = 1$  to  $N$  do
2:   for  $k = 1$  to  $K$  do
3:     Draw  $\theta_k^{(i)} \sim \text{Beta}(\alpha_k, \beta_k)$ 
4:   end for
5:   Compute  $\Theta^{(i)} = \frac{1}{K} \sum_k w_k \theta_k^{(i)}$ 
6: end for
7:  $\text{VaR}_\alpha \leftarrow \text{quantile}_\alpha(\{\Theta^{(i)}\}_{i=1}^N)$ 
8:  $\text{CVaR}_\alpha \leftarrow \mathbb{E}[\Theta \mid \Theta \leq \text{VaR}_\alpha]$ 
9: return  $\text{VaR}_\alpha, \text{CVaR}_\alpha$ , failure probability  $\hat{p} = \frac{1}{N} \sum_i \mathbf{1}[\Theta^{(i)} < \theta^*]$ 

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Figure 4 shows the results of a stress test across eight controls with heterogeneous posteriors, along with the pairwise failure correlation matrix that reveals which controls tend to fail together.

Figure 3: Monte Carlo Stress Testing of Control Portfolio

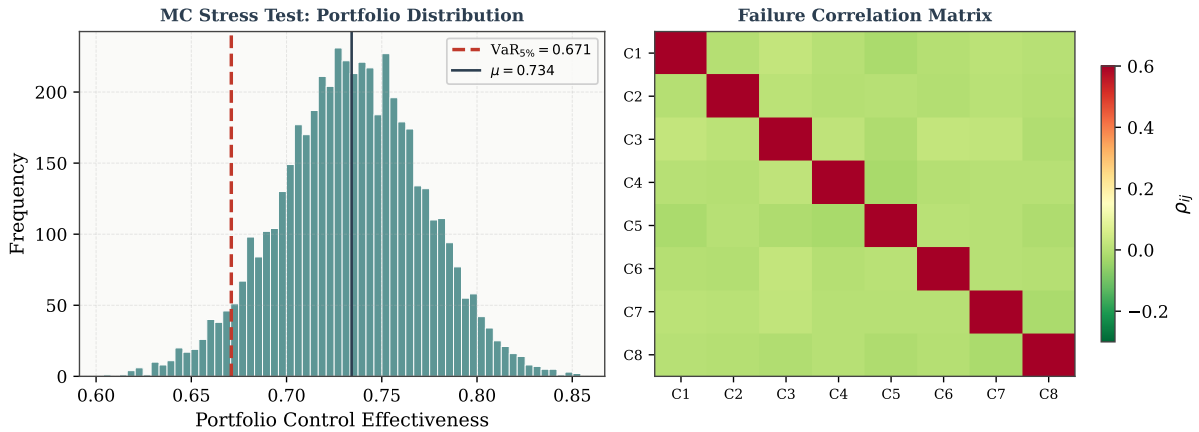


Figure 4: Monte Carlo stress test results. **Left:** Distribution of portfolio control effectiveness over 5,000 simulations, with 5% Value-at-Risk indicated. **Right:** Pairwise failure correlation matrix revealing co-failure structure.

5.3 Implications for Audit Planning

The VaR and CVaR estimates directly inform risk-based audit planning under ISA 300:

- Controls contributing most to tail risk receive priority in the next audit cycle.
- Correlated failures suggest shared root causes (e.g., common IT infrastructure, same process owner), directing the auditor toward systemic issues rather than isolated findings.
- The simulation can be conditioned on scenarios (“what if Control 3 degrades to 50% effectiveness?”) to perform sensitivity analysis.

6 Jensen’s Inequality and the Variance Penalty

Perhaps the most consequential theoretical result of the Bayesian governance framework concerns the interaction between *variance* in control effectiveness and *expected risk*.

Proposition 6.1 (Variance Penalty via Jensen’s Inequality). Let $R(\theta)$ be a convex risk function (e.g., $R(\theta) = -\ln \theta$, representing log-loss risk). Then for any random θ with $\text{Var}[\theta] > 0$:

$$\mathbb{E}[R(\theta)] \geq R(\mathbb{E}[\theta]), \quad (9)$$

with equality if and only if θ is deterministic. The gap $\mathbb{E}[R(\theta)] - R(\mathbb{E}[\theta]) > 0$ is the *Jensen gap*, and it grows with $\text{Var}[\theta]$.

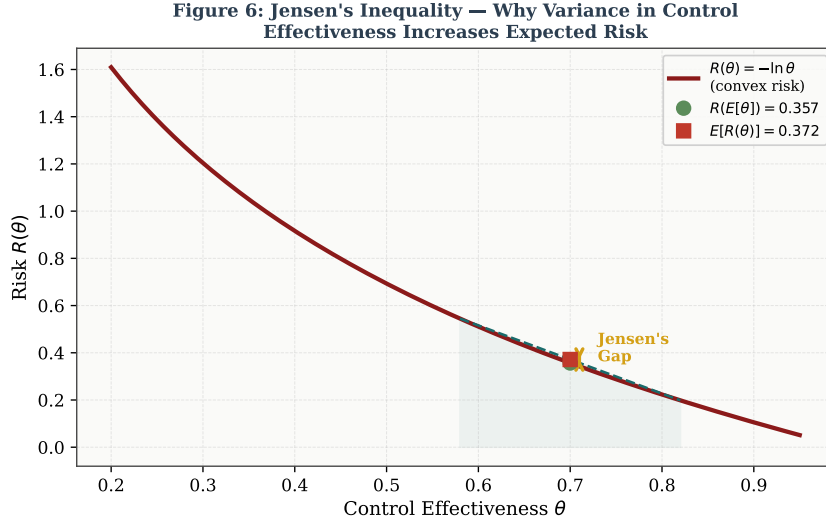


Figure 5: Jensen’s inequality applied to governance risk. The convex risk function $R(\theta) = -\ln \theta$ ensures that the expected risk $\mathbb{E}[R(\theta)]$ always exceeds the risk evaluated at the expected effectiveness $R(\mathbb{E}[\theta])$. Traditional audit approaches that report only point estimates systematically underestimate risk.

The implication is profound and direct: *any audit methodology that reports only point estimates of control effectiveness, without accounting for uncertainty, will systematically underestimate risk*. This is not a modelling artefact—it is a mathematical certainty whenever the risk function is convex, which it is for all standard risk metrics (log-loss, quadratic loss, exponential loss).

This result provides a rigorous justification for maintaining full posterior distributions rather than collapsing to point estimates, and it quantifies the “variance penalty” that organizations pay when they ignore uncertainty in governance assessments.

7 Integrated Framework Architecture

The components described in Section 2–Section 6 combine into an integrated continuous governance monitoring framework, depicted in Figure 6.

Figure 4: Bayesian Governance Framework — Conceptual Architecture

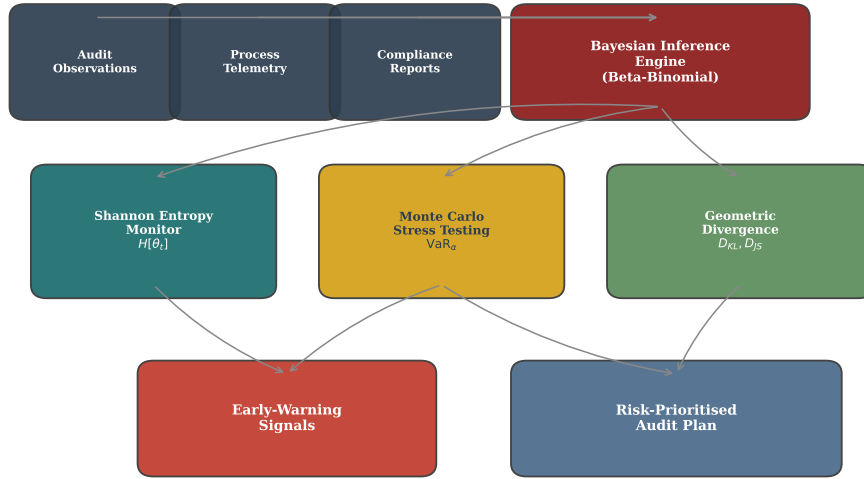


Figure 6: Conceptual architecture of the Bayesian Governance Framework. Audit observations and process telemetry feed a Beta-Binomial inference engine, which maintains calibrated posteriors. Three monitoring layers—entropy, Monte Carlo, and geometric divergence—produce early-warning signals and risk-prioritised audit plans.

The framework operates as a three-layer system:

1. **Data ingestion layer.** Audit observations (binary pass/fail from testing), process telemetry (continuous metrics from ERP/GRC systems), and compliance reports feed into the inference engine.
2. **Bayesian inference engine.** The Beta-Binomial conjugate model maintains a posterior $\text{Beta}(\alpha_t, \beta_t)$ for each control, updated in real time. For continuous telemetry, a discretisation step maps metric values to binary effectiveness indicators via pre-specified thresholds.
3. **Monitoring and alerting layer.** Three complementary monitors operate in parallel:
 - *Shannon entropy monitor:* detects posterior broadening indicative of regime change.
 - *Monte Carlo stress test:* propagates joint uncertainty to produce portfolio-level VaR/CVaR.
 - *Geometric divergence monitor:* tracks KL/JS distance from reference posteriors to detect distributional drift.
4. **Decision support layer.** Early-warning signals and stress-test results are synthesised into a risk-prioritised audit plan, updated continuously rather than annually.

8 Connections to Information Geometry and TDA

The Bayesian governance framework presented here is the *process-level* instantiation of a broader geometric programme developed across companion research papers.

8.1 Information Geometry (Paper A)

Paper A develops the full differential-geometric apparatus. The key insight is that the space of all Beta posteriors forms a Riemannian manifold $(\mathcal{M}, g_F)$ where g_F is the Fisher information metric (Equation (7)). On this manifold:

- Geodesics represent the most efficient “path” between two states of belief.
- Ricci curvature of the manifold at a point reflects the local *distinguishability* of nearby distributions—high curvature means small parameter changes produce large distributional shifts, and vice versa.
- The *Local Topological Deviation* (LTD) score, defined via persistent homology of the point cloud of posteriors over time, captures geometric deformation that precedes failure.

8.2 Topological Data Analysis (Papers A and C)

Persistent homology provides a multi-scale topological summary of the posterior trajectory. When a control’s effectiveness undergoes a regime change, the topology of the posterior point cloud in (α, β) -space changes: new homological features (loops, voids) appear in the persistence diagram. Paper C applies this machinery to specific audit domains—procurement (BPI 2019 data), R&D pipeline, and strategy execution signals from DEF 14A proxy filings—using SBERT embeddings and survival analysis.

8.3 Entity-Level Analysis (Paper B)

Paper B lifts the control-level analysis to the entity level, studying ten historical corporate failures through the lens of Bayesian Online Change-Point Detection (BOCPD) aligned with ISA 315 risk categories and Extreme Value Theory (EVT) aligned with ISA 300 materiality thresholds. The Beta-Binomial model from Section 2 serves as the building block: each entity’s governance trajectory is a time series of posteriors on the statistical manifold, and failure corresponds to crossing a critical boundary in this geometric space.

9 Discussion and Research Agenda

9.1 Summary of Contributions

This review has presented:

1. A **Beta-Binomial framework** for continuous governance monitoring that replaces binary audit judgments with calibrated posterior distributions (Section 2).
2. **Shannon entropy** as a principled early-warning signal for governance deterioration, with a leading-indicator property relative to the posterior mean (Section 3).
3. **KL and JS divergence** for formal regime-change detection on the statistical manifold of Beta distributions (Section 4).
4. **Monte Carlo stress testing** for portfolio-level risk quantification with VaR/CVaR estimates (Section 5).
5. A **Jensen’s inequality result** proving that point-estimate-based audit methodologies systematically underestimate risk (Section 6).
6. An **integrated architecture** combining all components into a continuous monitoring system (Section 7).

9.2 Limitations

The Beta-Binomial model assumes exchangeable binary observations, which may not hold when control effectiveness exhibits strong serial correlation or seasonality. Extensions to Hidden Markov Models (HMMs) with Beta emission distributions, developed in Paper B, address this limitation. The independence assumption across controls in the Monte Carlo stress test (Algorithm 1) can be relaxed using copula models or vine copulas to capture joint tail dependence.

9.3 Open Problems and Future Directions

1. **Adaptive priors.** Power priors or dynamic linear models that allow α_0, β_0 to evolve over time, discounting stale observations.
2. **Multi-level models.** Hierarchical Beta-Binomial models where controls within the same process share a common hyper-prior, enabling information sharing across related controls.
3. **Causal structure.** Integrating Bayesian networks to model causal dependencies between controls, so that failure of an upstream control propagates correctly to downstream risk estimates.
4. **Regulatory alignment.** Mapping the framework’s outputs to EU AI Act conformity assessment requirements (Article 9, risk management; Article 17, quality management) and FDA 21 CFR Part 11 for pharmaceutical applications.
5. **Topological persistence.** Full development of the persistent-homology-based governance monitoring, including stability results connecting topological features of the posterior trajectory to audit-relevant risk categories (Paper A).
6. **Empirical validation.** Application to real-world audit data (BPI 2019 procurement logs, CMS Medicare billing data) to validate the entropy early-warning property and divergence-based regime detection against known failure events.

9.4 Concluding Remarks

The transition from periodic, binary audit to continuous, probabilistic governance monitoring is not optional—it is being mandated by regulation (EU AI Act), enabled by technology (real-time ERP telemetry, cloud-native GRC platforms), and demanded by the complexity of modern enterprise risk landscapes. The mathematical tools presented here—Bayesian inference, information theory, statistical geometry—are not new. What is new is their systematic application to the governance domain, and the recognition that audit assurance is, at its core, a problem of *statistical inference on manifolds*.

This research programme aims to make this recognition operational: providing auditors with rigorous, interpretable, and computationally tractable tools for continuous assurance in regulated industries.

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